

Address Decoding

Memory Space		A[11]	A[10]	A[9]	A[8]	CS_ROM	CS_RAM
		A	B	C	D		
ROM	0x000	0	0	0	0	0	1
	0xBFF	1	0	1	1	0	1
RAM	0xC00	1	1	0	0	1	0
	0xEFF	1	1	1	0	1	0
MMIO	0xF00	1	1	1	1	1	1
	0xFFF	1	1	1	1	1	1

For ROM: CS = 0 on 0000 to 1011, All Others: CS = 1

	CD AB	00	01	11	10
00		0 0	0 1	0 3	0 2
01		0 4	0 5	0 7	0 6
11		1 12	1 13	1 15	1 14
10		0 8	0 9	0 11	0 10

$$\text{ROM_CS} = A \cdot B$$

To do this with nand only we can nand A, B then take the output and feed it into both inputs of another nand gate to act like an inverter, creating an AND

For RAM: CS = 0 on 1100 to 1110, All Others: CS = 1

	CD AB	00	01	11	10
00		1 0	1 1	1 3	1 2
01		1 4	1 5	1 7	1 6
11		0 12	0 13	1 15	0 14
10		1 8	1 9	1 11	1 10

$$\text{RAM_CS} = A' + B' + CD$$

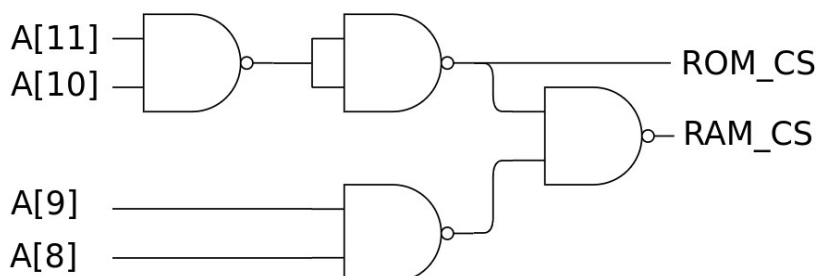
To do this with nand, we first apply DeMorgan's

$$\text{RAM_CS} = (A \cdot B \cdot (C \cdot D)')'$$

$(C \cdot D)'$ is the same as nand; i.e. $\text{nand}(C, D)$ while $A \cdot B = \text{ROM_CS}$ lets reuse it...

$$\text{RAM_CS} = (\text{ROM_CS} \cdot (C \cdot D)')' = \text{nand}(\text{ROM_CS}, \text{nand}(C, D))$$

$$\text{ROM_CS} = A[11] \cdot A[10] \quad \text{RAM_CS} = (\text{ROM_CS} \cdot (A[11] \cdot A[10])')'$$



Note: 74LVC00

Propagation Delay = 4.1ns

Worst Case for 3 Gate Path
Total = 3(4.1ns) = **12.3ns**